

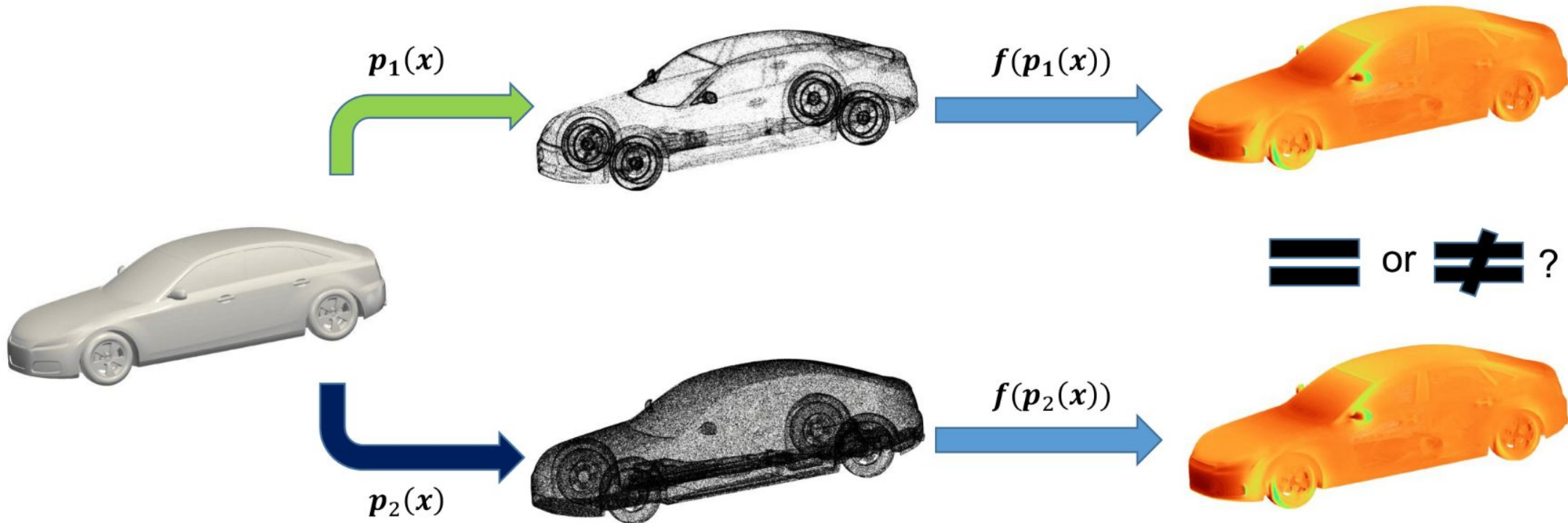
Meeting 17.06.2026

Update meeting with Parsa

Chapter 1: Is GMP/SMART sample agnostic or point sampling strategy makes a noticeable change in predictions and accuracy

Motivation

- When we are making prediction y for an input geometry x , our prediction is expected to be $f(x) = y$. However, geometries are first preprocessed into a pointcloud using $p(x)$ and then they are fed to the model. So what we get is actually $f(p(x))$. Not having access to the $p(x)$ poses a challenge for an ideal prediction if the network is not agnostic to how behaves.
- We need to test whether for two different sampling strategy $p_1(x)$ and $p_2(x)$ or not.

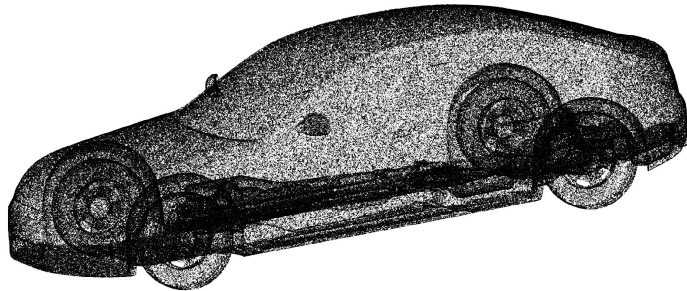


Testing GMP (Geometry pretrained + 500 DrivAerML Physics trained)

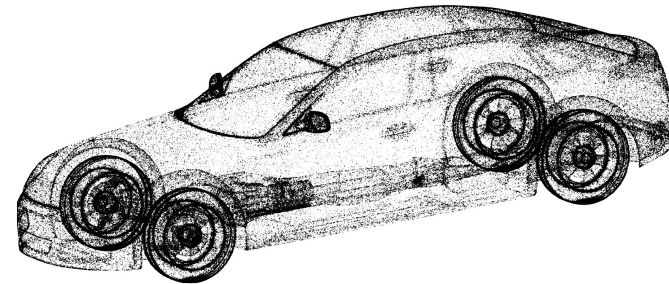
Neglecting the CFD results at this slide, we just focus on seeing the GMP predictions on the sampling strategy that came with DrivAerML vs. The sampling strategy from the Audi (using their own CAD tool to re-mesh the low-quality stl file)

DrivAerML had access to the high quality stl but published a compressed low triangle version. For this reason we had to re-mesh the stl to be able to run it in CFD. This re-meshing created a difference in the distribution of vertices.

For example both of these show 0.5M vertices, while the distribution is clearly different:

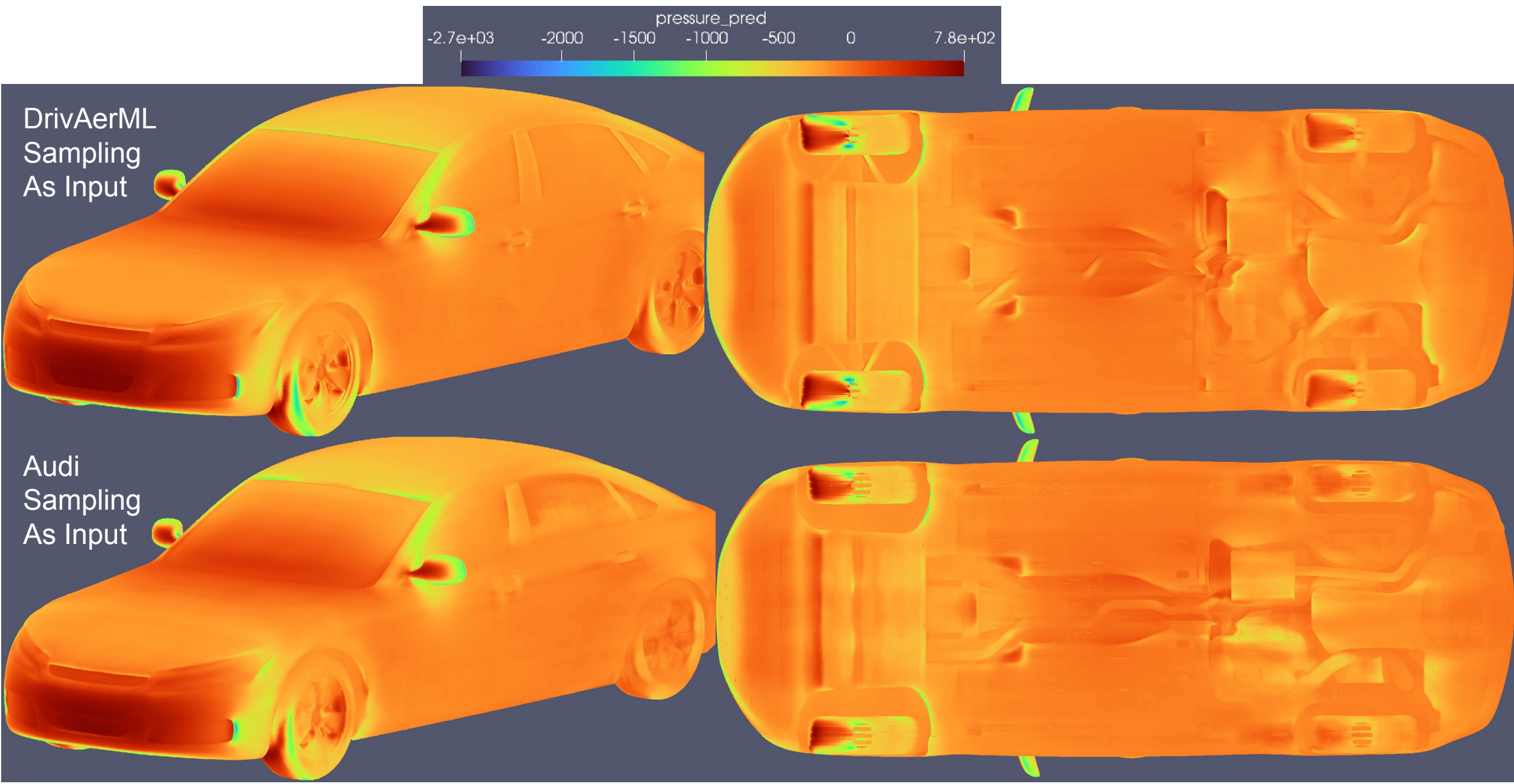


Audi Meshing Strategy



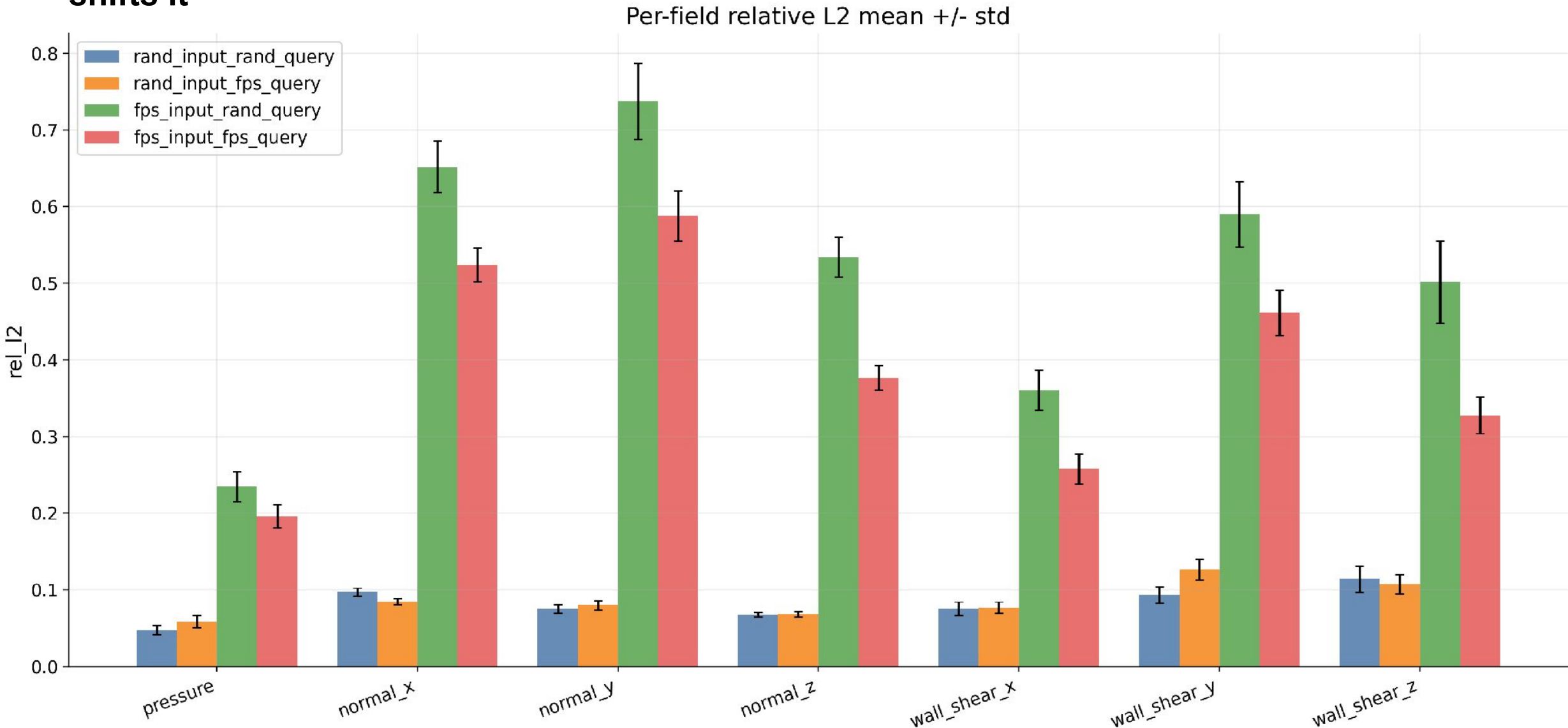
Equivalebt DrivAerML Sample

Testing GMP (Geometry pretrained + 500 DrivAerML Physics trained) - Predictions



Four Experiments to see the effect of distribution in both input and query on error
On GMP that has been trained on random sub-sampling for both input and query

The point distribution here is from DrivAerML, random sub-sampling preserves it while FPS shifts it



The mathematics behind the issue

In an anchored attention-based encoder, we aggregate the features of the input point cloud using the below mentioned equation. This features can be the position embedding or extra attributes like surface normals:

$$z_i = \frac{\int_S \exp(s_i(x)) v(x) \rho(x) dA(x)}{\int_S \exp(s_i(x)) \rho(x) dA(x)}$$

where:

- S : true continuous surface,
- x : point on the surface,
- $dA(x)$: true surface-area element,
- $\rho(x)$: sampling density,
- $s_i(x)$: attention score from latent anchor i to surface point x ,
- $v(x)$: value/features at point x ,
- z_i : aggregated latent feature.

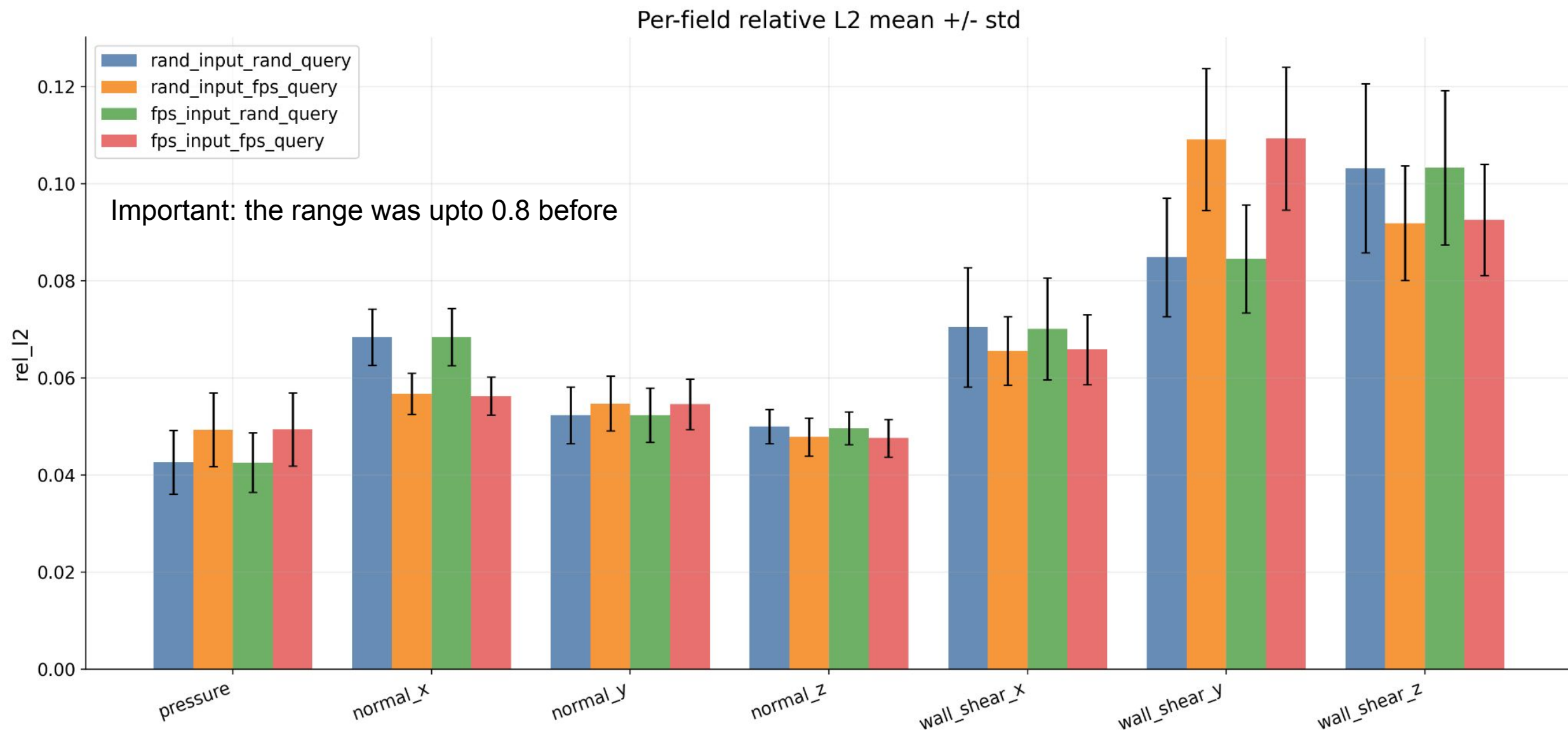
The mathematics behind the solution

$$z_i = \frac{\int_S \exp(s_i(x)) v(x) \rho(x) dA(x)}{\int_S \exp(s_i(x)) \rho(x) dA(x)} \xrightarrow{s_i^{\text{corrected}} = s_i - \log \hat{\rho}_i} \frac{\int_S \exp(s_i(x)) v(x) dA(x)}{\int_S \exp(s_i(x)) dA(x)}$$

where:

- S : true continuous surface,
- x : point on the surface,
- $dA(x)$: true surface-area element,
- $\rho(x)$: sampling density,
- $s_i(x)$: attention score from latent anchor i to surface point x ,
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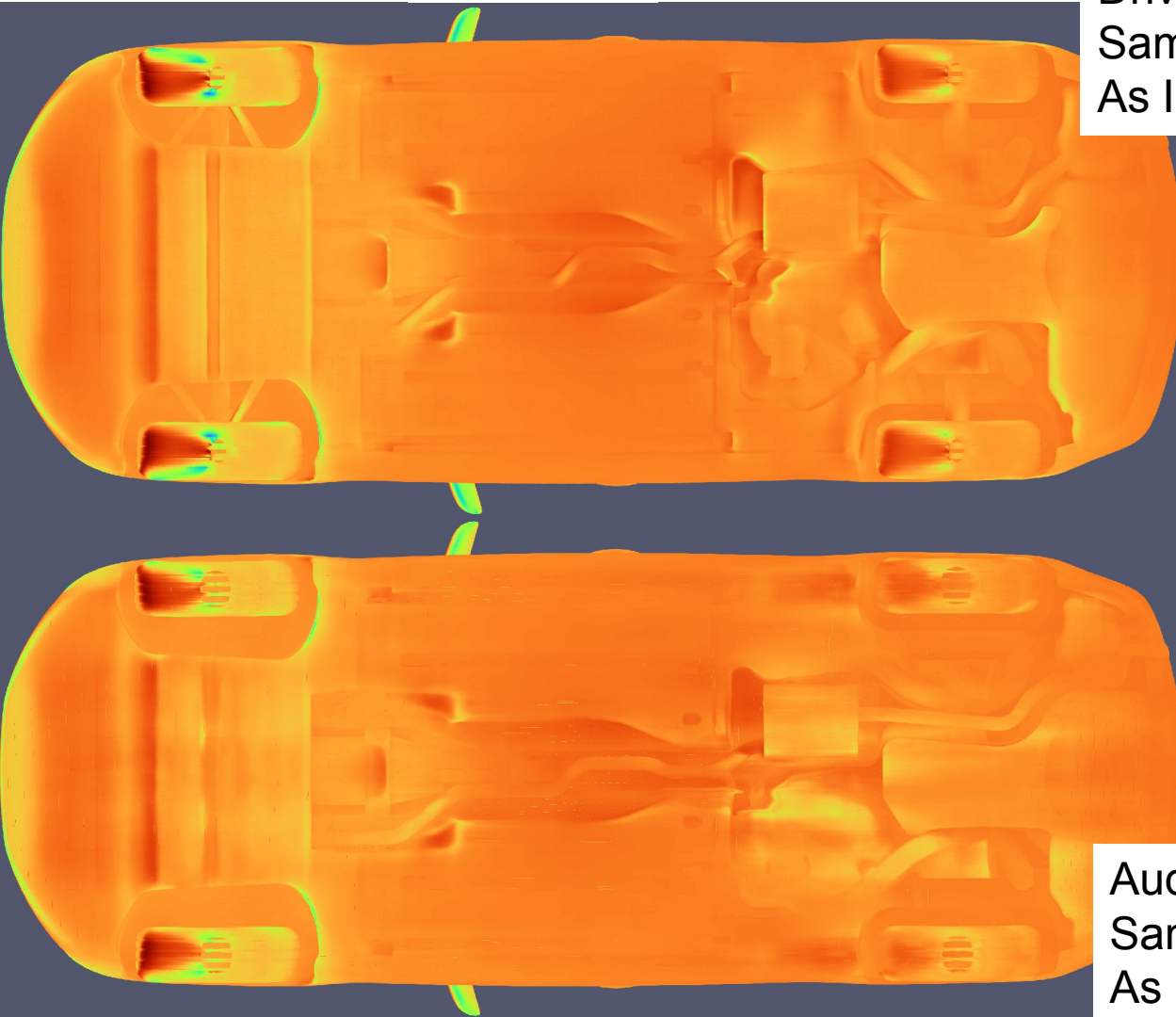
Result after the solution (on 25 test subsample)



GMP output comparison after density correction

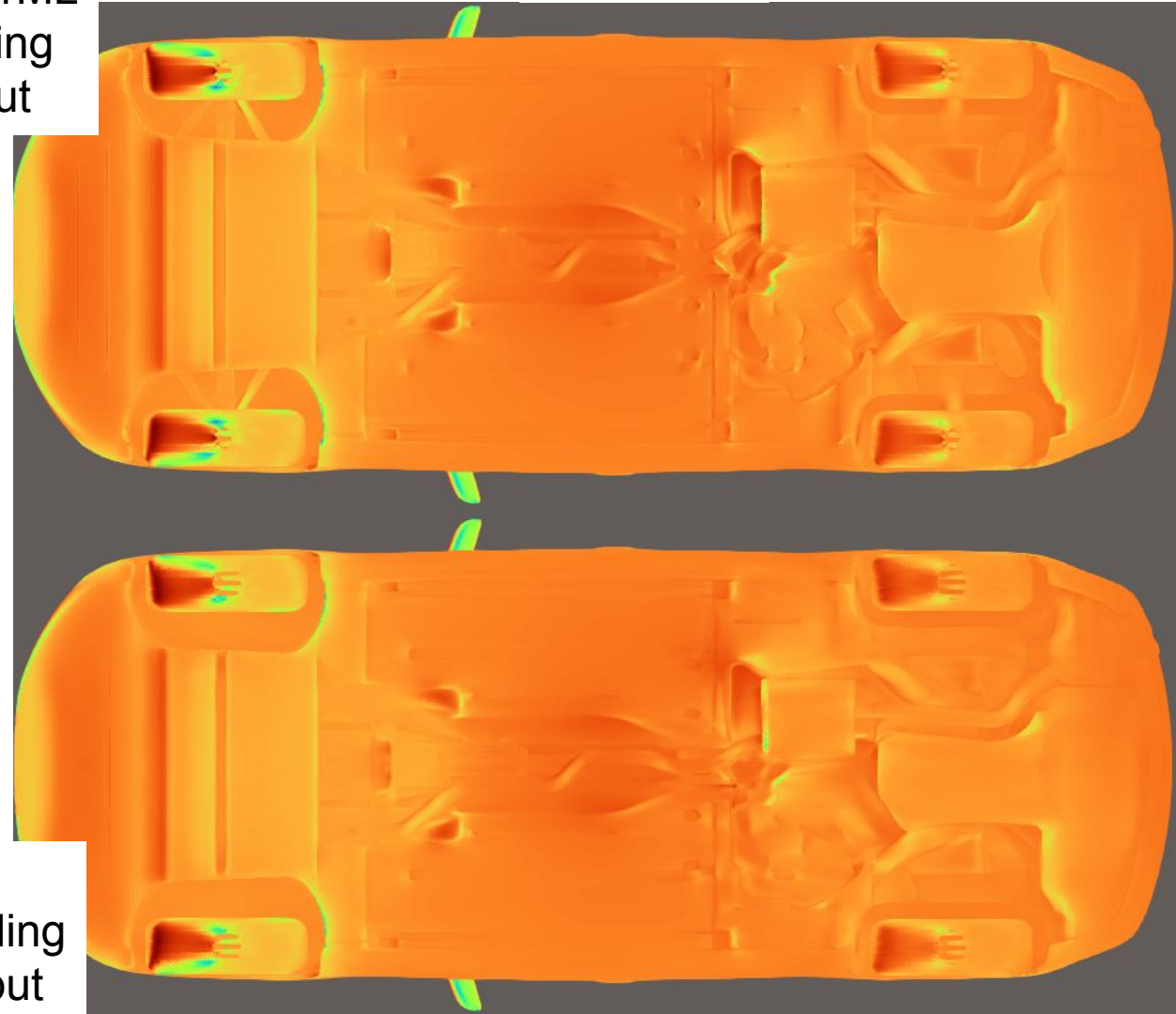


Before



DrivAerML
Sampling
As Input

After

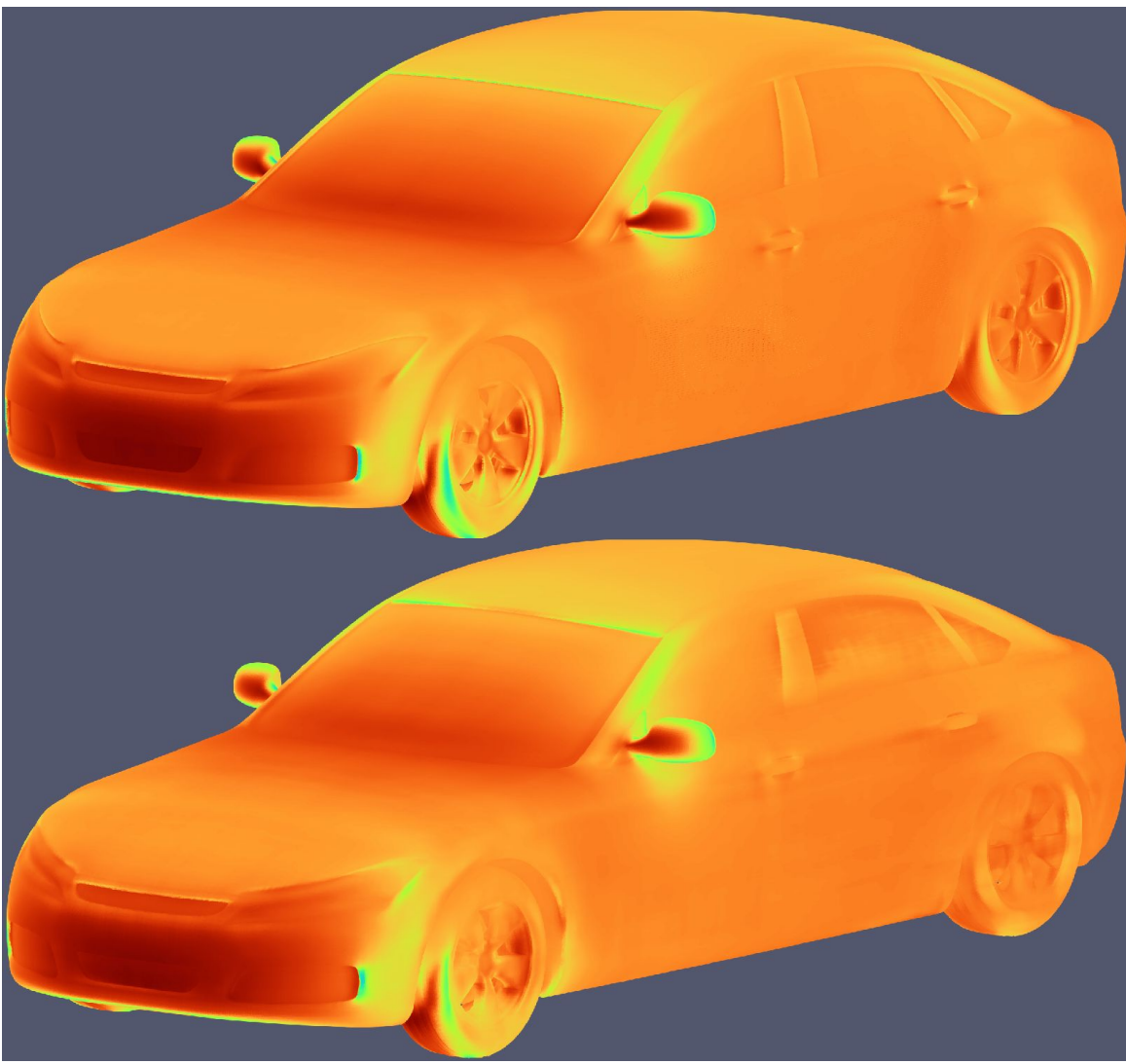


Audi
Sampling
As Input

GMP output comparison after density correction



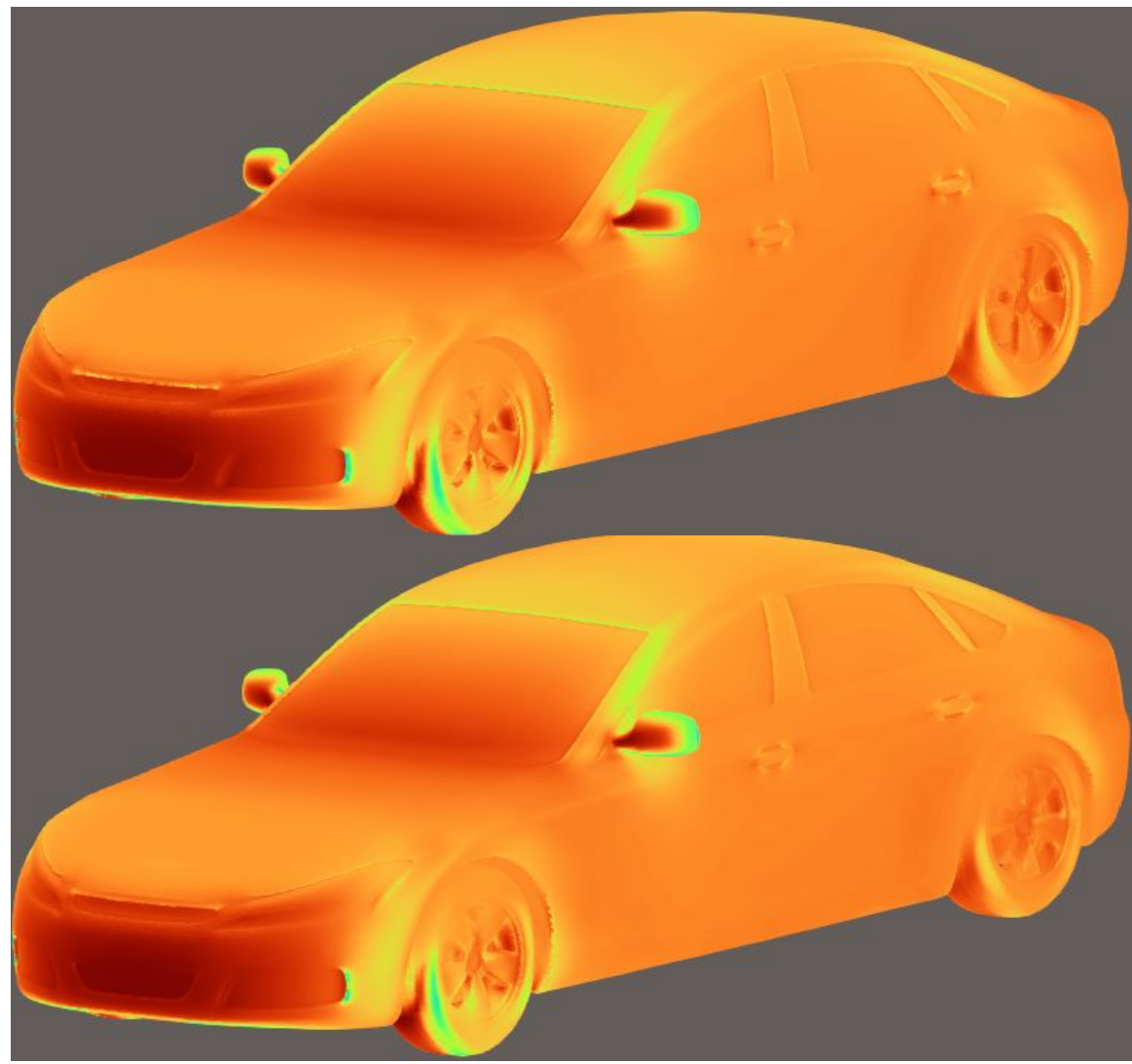
Before



DrivAerML
Sampling
As Input

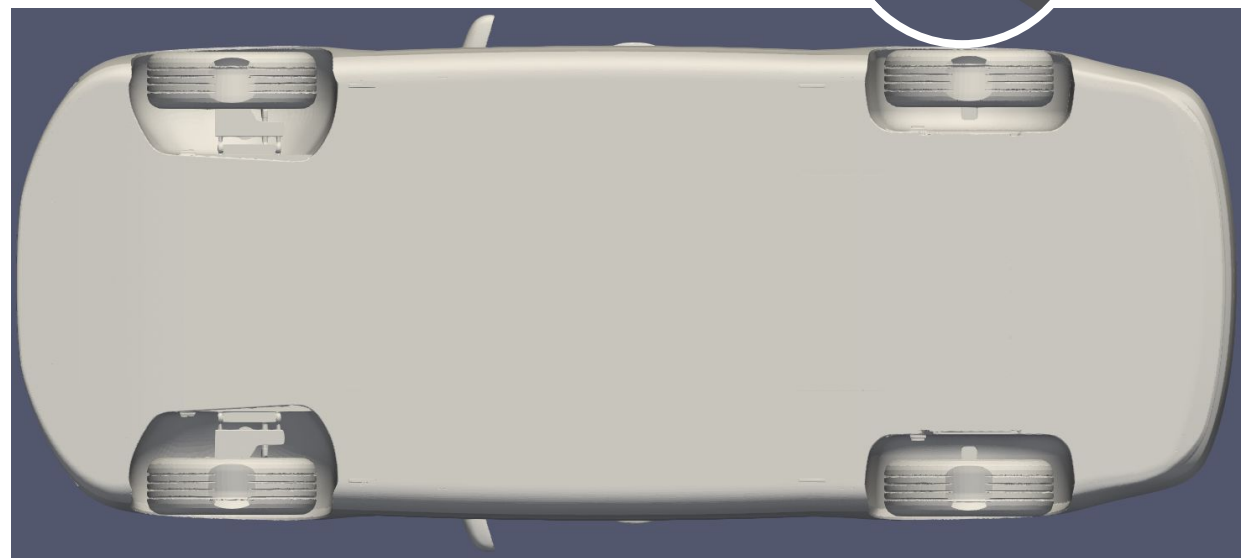
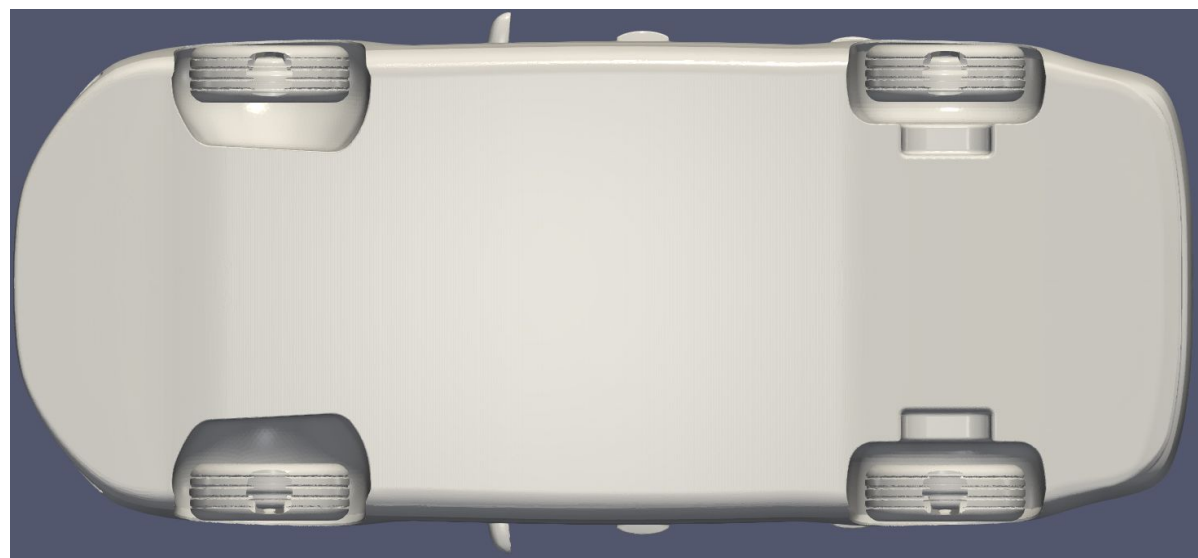
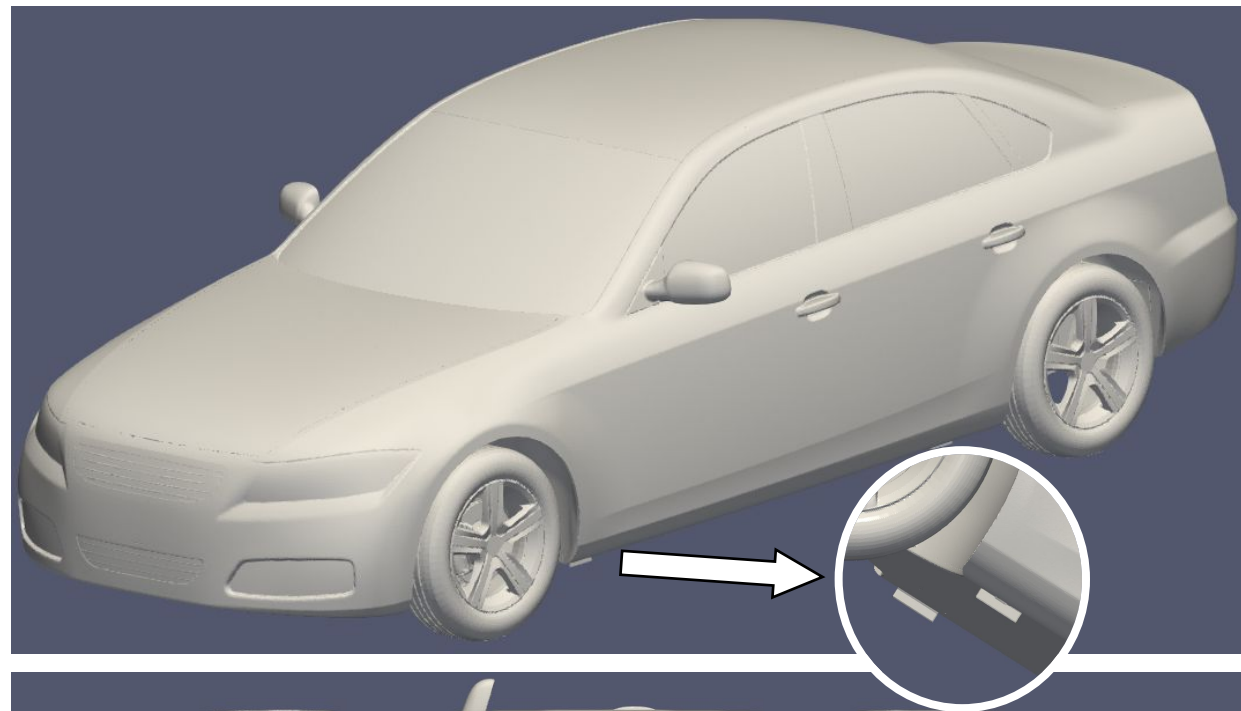
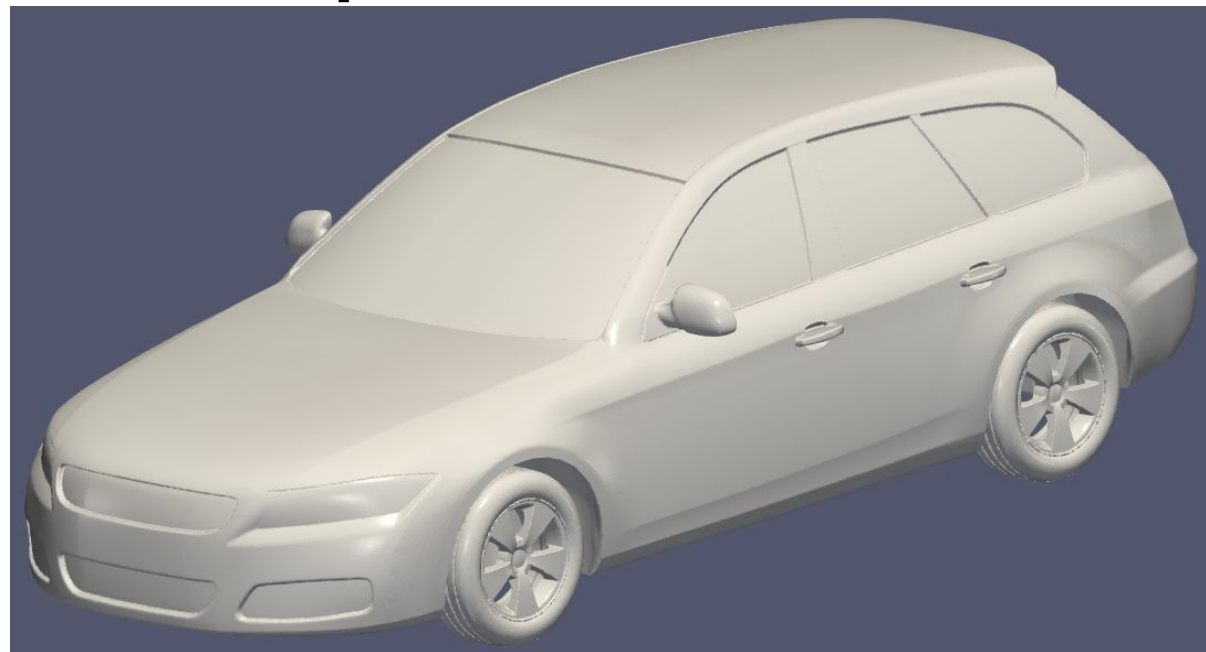
Audi
Sampling
As Input

After

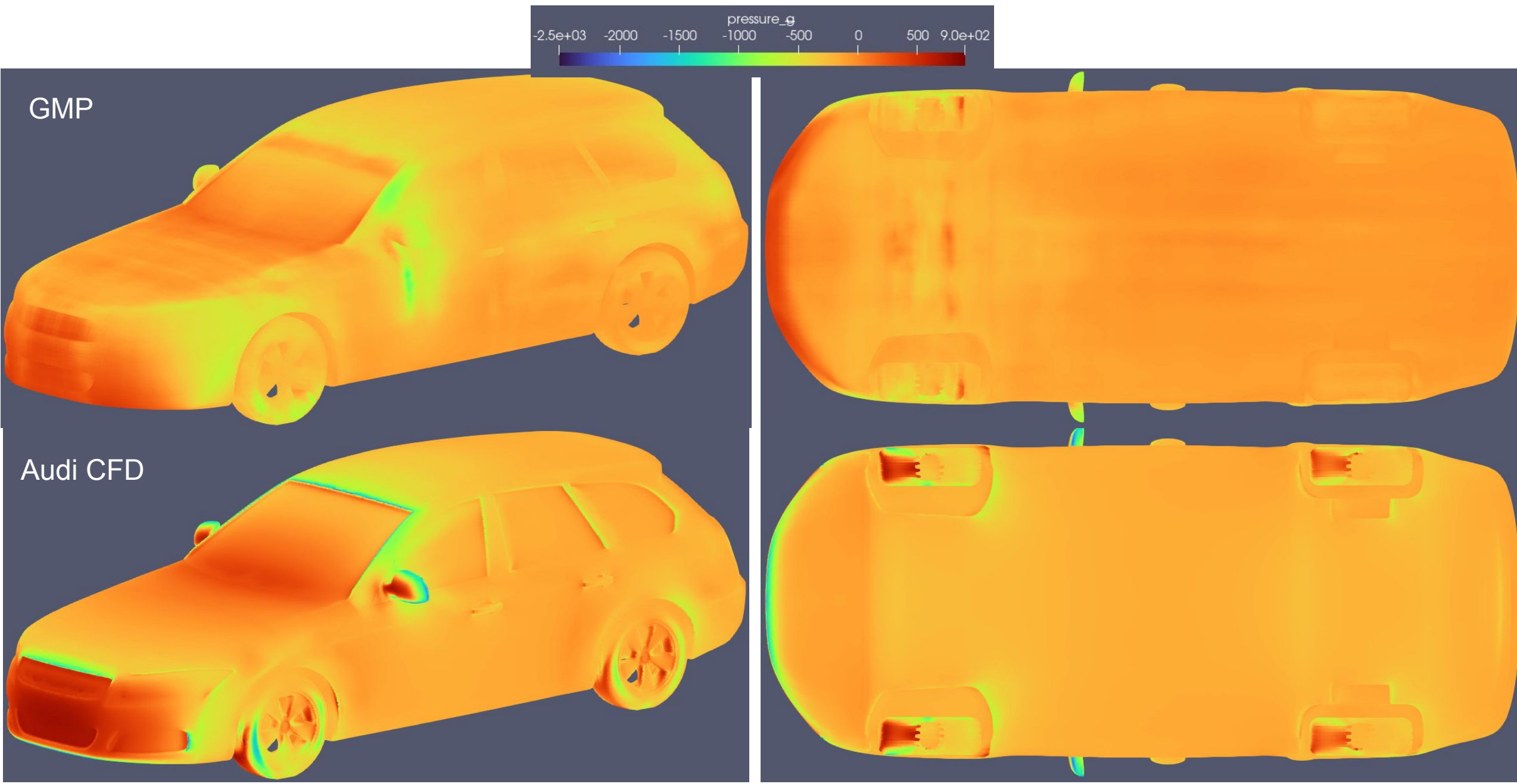


Chapter 2: Trying further geometries with Audi CFD and comparing the prediction of GMP with this ground truth

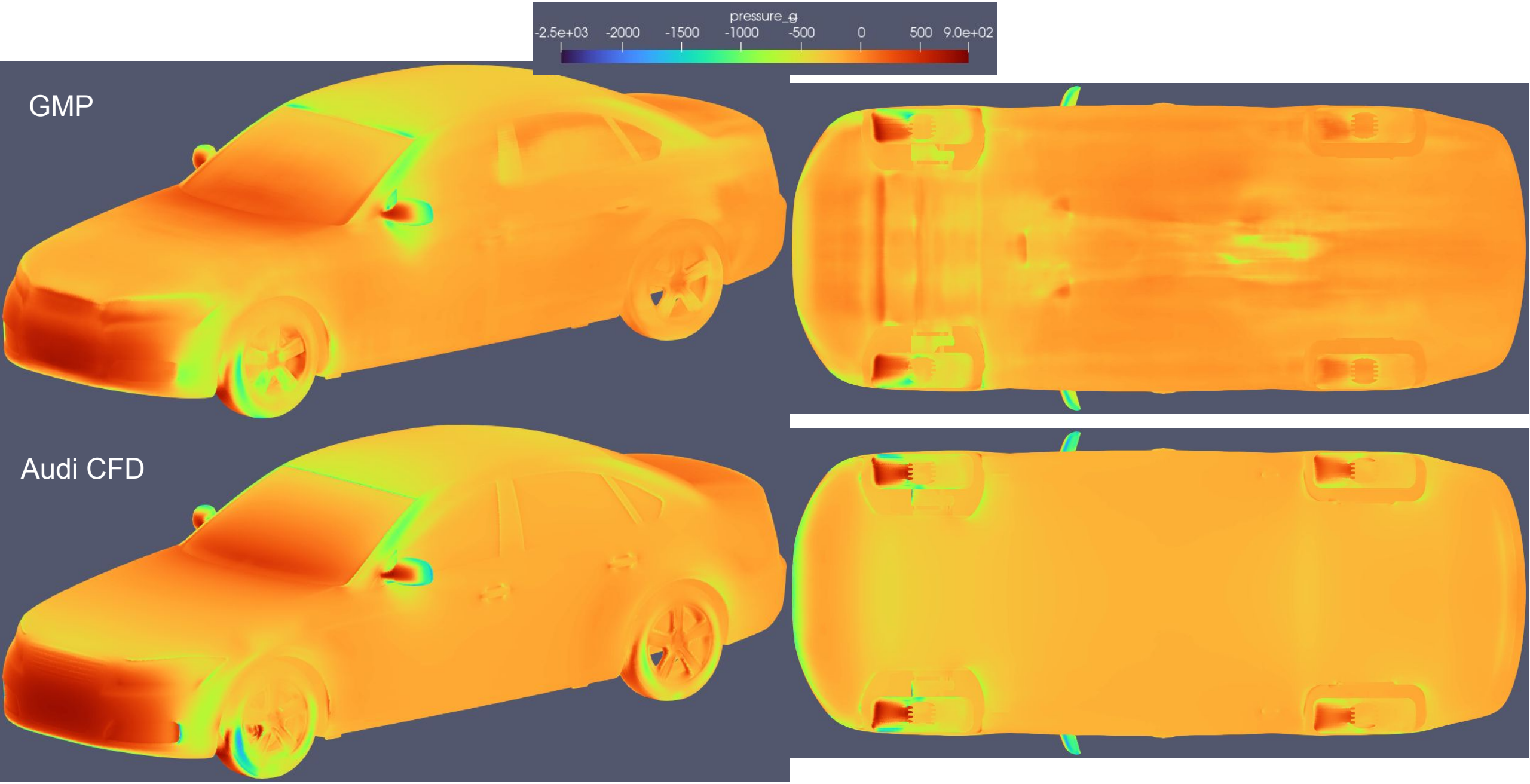
Experimented Geometries



Audi CFD results vs GMP prediction - Estate-back



Audi CFD results vs GMP prediction – Notchback



Chapter 3: Chapter 1 + Chapter 2

